

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0891

Still Mill 138 kV
October 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0891. The Interconnection Customer proposes a 100.8 MW wind facility interconnecting at the Still Mill 138 kV substation in the Virginia area. The request was received on November 13, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,632,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Cia - Papago Buttes 138 KV
Contingency: Cia - Papago Buttes 138 KV Line
Loading: 111.9% (Rating: 286 MVA)
DFAX: 9.3%
Pre-project loading: 108.6%
2. Facility: Mottman - Papago Buttes 138 KV Ckt 1
Contingency: Papago Buttes - Mottman 138 KV Cir. 2
Loading: 111.1% (Rating: 321 MVA)
DFAX: 9.7%
Pre-project loading: 108.0%
3. Facility: Papago Buttes - Mottman 138 KV Cir. 2
Contingency: Papago Buttes - Mottman 138 KV Line 1
Loading: 111.1% (Rating: 321 MVA)
DFAX: 9.7%
Pre-project loading: 108.0%
4. Facility: Cia - New Sterling 138 KV
Contingency: Cia - Cowlitz Falls 138 KV
Loading: 110.4% (Rating: 258 MVA)
DFAX: 9.7%
Pre-project loading: 106.6%
5. Facility: Melba - Still Mill 138 KV Line
Contingency: Cmanche Peak 345/1kV Transformer
Loading: 107.3% (Rating: 204 MVA)
DFAX: 44.0%
Pre-project loading: 85.5%

6. Facility: Cia - Papago Buttes 138 KV line
 Contingency: Papago Buttes - Cia 138 KV Ckt 2
 Loading: 106.4% (Rating: 301 MVA)
 DFAX: 9.3%
 Pre-project loading: 103.3%
7. Facility: Mottman - Bloomington 138 KV Cir. 1
 Contingency: Mottman Substation - Bloomington 138 KV
 Loading: 104.4% (Rating: 419 MVA)
 DFAX: 16.8%
 Pre-project loading: 100.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Papago Buttes - Cia 138 KV Line 2	Rebuild line with higher-capacity conductor	\$2,440,000	42% (\$1,021,000)
Papago Buttes - Mottman 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$1,601,000	13% (\$211,000)
Papago Buttes - Mottman 138 KV Cir. 2	Rebuild line with higher-capacity conductor	\$1,212,000	13% (\$160,000)
Cia - New Sterling 138 KV line	Rebuild line with higher-capacity conductor	\$1,632,000	13% (\$215,000)
Melba - Still Mill 138 KV Line	Reconductor line	\$2,386,000	98% (\$2,337,000)
Cia - Papago Buttes 138 KV line	Reconductor line	\$881,000	42% (\$369,000)
Mottman - Bloomington 138 KV Cir. 1	Reconductor line	\$592,000	54% (\$319,000)
	Total allocated to project		\$4,632,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.